

House Price Prediction Using Machine Learning

Summary Report

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, parking availability, air conditioning, furnishing status, and location-related attributes. The project also aimed to identify the factors that have the greatest influence on house prices and compare the performance of different regression algorithms.

Dataset

The project used the **Housing Prices Dataset** from Kaggle, containing residential property information and corresponding house prices. The dataset includes numerical and categorical features such as area, bedrooms, bathrooms, stories, parking spaces, main road access, guest room availability, basement, hot water heating, air conditioning, preferred area, and furnishing status.

Data Preparation

The dataset was loaded using the Pandas library and examined for missing values and duplicate records. Missing values were removed (if any), duplicate rows were eliminated, and categorical variables were converted into numerical format using one-hot encoding. This preprocessing ensured that the dataset was suitable for machine learning algorithms.

Models Used

Two regression models were trained and evaluated:

- **Linear Regression**
- **Random Forest Regressor**

The dataset was divided into training (80%) and testing (20%) sets. Model performance was evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Results

Both models were able to predict house prices successfully. However, the **Random Forest Regressor** achieved better overall performance, with a higher R^2 score and lower prediction errors compared to the Linear Regression model. This indicates that Random Forest captured more complex relationships between property features and house prices.

The visualizations, including the house price distribution, correlation heatmap, and actual versus predicted price scatter plot, helped explain the dataset and evaluate model performance.

Key Findings

The analysis showed that **area** was the most influential factor affecting house prices. Other important features included the number of bathrooms, parking spaces, air conditioning, stories, and preferred area. Properties with larger living areas and additional amenities generally had higher market values.

Recommendation

Real estate businesses can improve price estimation by using machine learning models instead of relying only on manual comparisons. Property features such as area, parking, bathrooms, and modern amenities should receive greater attention during property valuation. A Random Forest model can provide more reliable price predictions and support better investment and pricing decisions.

Conclusion

This project demonstrated how machine learning can be applied to predict house prices accurately using property characteristics. Proper data preprocessing, feature engineering, and model evaluation significantly improved prediction quality. Among the models tested, the Random Forest Regressor produced the most accurate results, making it a suitable choice for practical house price prediction applications.